

REMOTE CONTROLLED CAR USER INFORMATION

1. What is the application and what is its Purpose?

The purpose of the application is to function like a remote control for a toy car, similar to the ones we enjoyed during childhood—and perhaps even as adults—but instead of using a traditional remote, you can control it directly from your phone.

2. How to Run the Application on Raspberry Pi?

This application has two parts: a mobile application and a Raspberry Pi component. You can download the mobile app either from GitHub or from the following link:

https://dosya.co/hnawrfqgyv43/remote_control_mobile_car.apk.html

I will explain the mobile part using the link above. For now, let's focus on the Raspberry Pi setup.

You need to SSH into your Raspberry Pi or use another method to transfer the `remote_control_mobile_car.py` file into your Raspberry Pi. Along with this file, also transfer the `requirements.txt` file. Then, run the following command to install the necessary dependencies:

```
pip install -r requirements.txt --break-system-packages
```

If you are unfamiliar with Raspberry Pi or how to ensure that the Raspberry Pi starts up automatically, you can refer to the following guide for detailed instructions:

<https://github.com/Acrome-Smart-Motion-Devices/Acrome-Motion-Kit-AI-Control-Global-/blob/main/Acrome%2BGlobal%20Motion%20Device%20Control%20.pdf>

Once you follow the steps in the guide, the Raspberry Pi setup should be complete.

This application works in both Hotspot mode and WiFi mode—whether both devices are connected to the same network, or if the Raspberry Pi has opened its own Hotspot and your phone connects to it. In other words, the application does not require internet access to function.

3. Application (Mobile Side)

When you launch the app, the first screen you will encounter is the IP Scanner screen. By clicking the "Scan the Devices" button, you can view the subnet IP address and all Linux devices connected to the network. (Although it could detect all devices, it only shows Linux devices because the Raspberry Pi uses a Linux-based operating system, so only relevant IP addresses are displayed.)

After selecting the IP address, a Continue button will appear. Once you click Continue, you will be directed to the main screen.

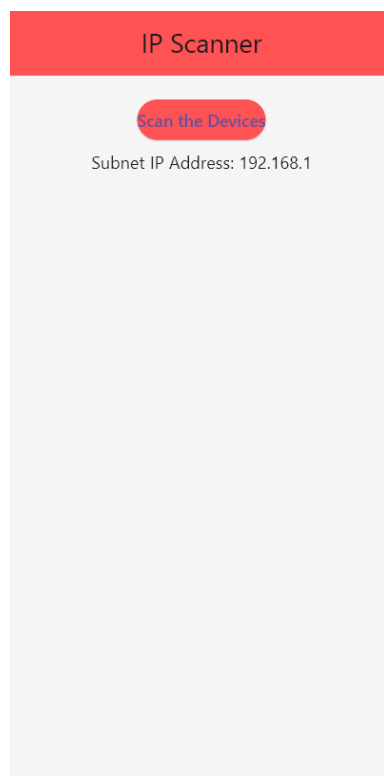


Figure 1: IP Scanner

The main screen is simple and intuitive. You can control the robot's movement using the arrow buttons. However, there is a small caution: if you press the Forward and Turn Right buttons simultaneously, the robot might get stuck in one direction. If this happens, exit the app and re-enter it. Then, press the Stop button or any direction button to stop the robot.

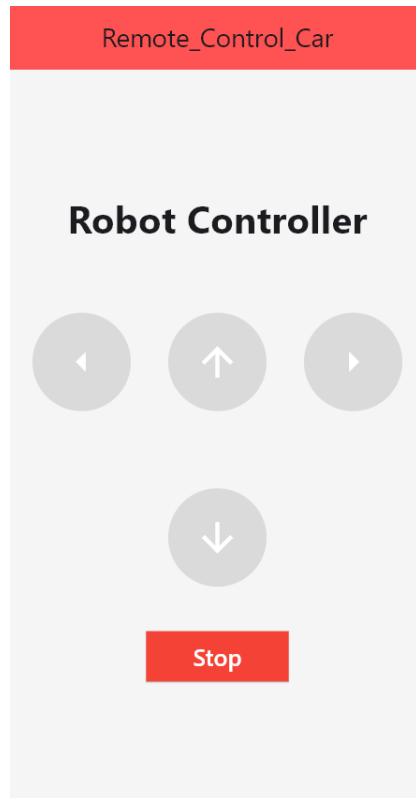


Figure 2: Main Screen